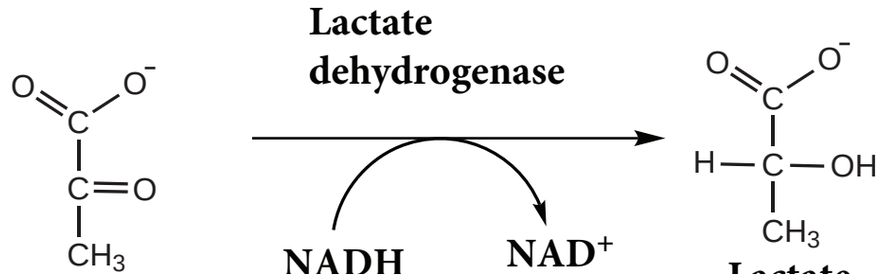
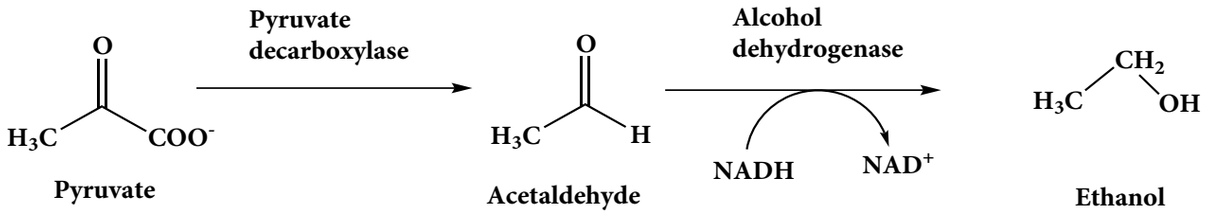


Many yeasts convert glucose to ethanol and CO_2 rather than pyruvate using alcoholic fermentation process. In this process, pyruvate is first converted to acetaldehyde by enzyme pyruvate decarboxylase in the presence of thiamine pyrophosphate and Mg^{2+}. Carbon dioxide is released during this reaction. Acetaldehyde is then converted to ethanol by the enzyme alcohol dehydrogenase during which the NADH is oxidized to NAD^+.



3.4. TRICARBOXYLIC ACID CYCLE:

Citric acid cycle is the common mode of oxidative degradation in eukaryotes and prokaryotes. This process occurs in the mitochondria. This cycle, which is also known as the tricarboxylic acid (TCA) cycle or Krebs cycle, is the central driver of cellular respiration. It accounts for the major part of oxidation of carbohydrate, fatty acid & amino acid and it also generates many biosynthetic precursors. The citric acid cycle is amphibolic, i.e. it operates both catabolically and anabolically.

Unlike glycolysis, the citric acid cycle is a cyclic process. i.e. the last reaction of the pathway regenerates the compound utilised in the first reaction. The citric acid cycle produces only one GTP molecule (ATP equivalent) directly and does not directly consume oxygen.

3.4.1. Reactions of tricarboxylic acid cycle

TCA cycle takes acetyl CoA produced by the oxidation of pyruvate (derived from glucose and other catabolic processes) as its starting material and in a series of reactions, harvests much of its bond energy in the form NADH, $FADH_2$ and GTP molecules. The reduced electron carriers NADH and $FADH_2$ generated in the TCA cycle, will pass their electrons into the electron transport chain resulting in oxidative phosphorylation, which will generate most of the ATP produced in cellular respiration.

Prior to the start of the first step, a transitional phase occurs during which pyruvate is converted to acetyl CoA. This process occurs in the mitochondrial matrix and provides a link between the glycolysis and the TCA Cycle which consists the following eight steps.

XII U3 Carbohydrates.indd 71 4/8/2020 3:06:52 PM